

# Assignment: Practical - 4 : Wireshark

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**Sem:** 2

**Subject:** Network Security

**Course:** MSc Cyber Security

# PART - A

## Task 1 - ICMP

1. Start Wireshark. Select your lab interface (e.g., eth0).
2. Generate ICMP Traffic by running:  
*nmap -sn <target>*
3. Apply the following display filters:  
*icmp*  
*icmp.type == 8*  
*icmp.type == 0*

```
(aastha@kali)-[~]
$ nslookup scanme.nmap.org
Server:      14.139.5.5
Address:     14.139.5.5#53

Non-authoritative answer:
Name:   scanme.nmap.org
Address: 45.33.32.156
Name:   scanme.nmap.org
Address: 2600:3c01::f03c:91ff:fe18:bb2f
```

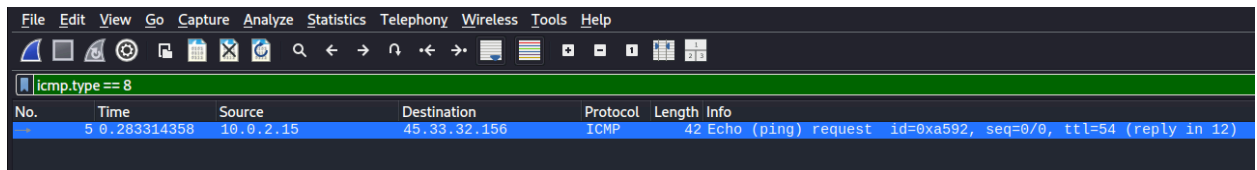
```
(aastha@kali)-[~]
$ nmap -sn scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-02 04:39 -0500
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00068s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 0.52 seconds
```

### Analyse

1. What is the total number of ICMP packets captured? Ans: 3

| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help |             |              |              |          |        |                                                              |
|----------------------------------------------------------------------------|-------------|--------------|--------------|----------|--------|--------------------------------------------------------------|
| icmp                                                                       |             |              |              |          |        |                                                              |
| No.                                                                        | Time        | Source       | Destination  | Protocol | Length | Info                                                         |
| 5                                                                          | 0.283814358 | 10.0.2.15    | 45.33.32.156 | ICMP     | 42     | Echo (ping) request id=9xa592, seq=0/0, ttl=54 (reply in 12) |
| 8                                                                          | 0.283955132 | 10.0.2.15    | 45.33.32.156 | ICMP     | 54     | Timestamp request id=0xd889, seq=0/0, ttl=43                 |
| 12                                                                         | 0.657360569 | 45.33.32.156 | 10.0.2.15    | ICMP     | 60     | Echo (ping) reply id=0xa592, seq=0/0, ttl=255 (request in 5) |

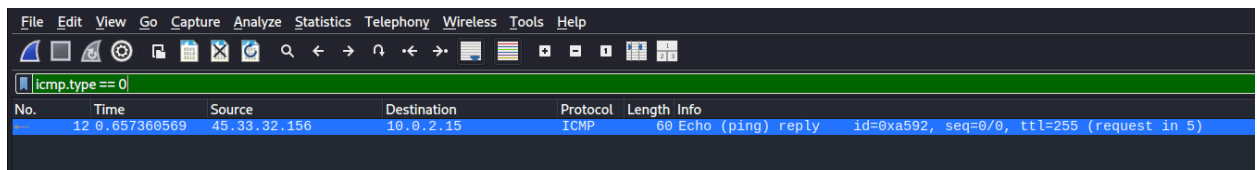
2. How many ICMP Echo Requests were sent? (Use filter icmp.type == 8) Ans: 1



The screenshot shows the Wireshark interface with the filter 'icmp.type == 8' applied. A single packet is listed in the packet list pane.

| No. | Time        | Source    | Destination  | Protocol | Length | Info                                                         |
|-----|-------------|-----------|--------------|----------|--------|--------------------------------------------------------------|
| 5   | 0.283314358 | 10.0.2.15 | 45.33.32.156 | ICMP     | 42     | Echo (ping) request id=0xa592, seq=0/0, ttl=54 (reply in 12) |

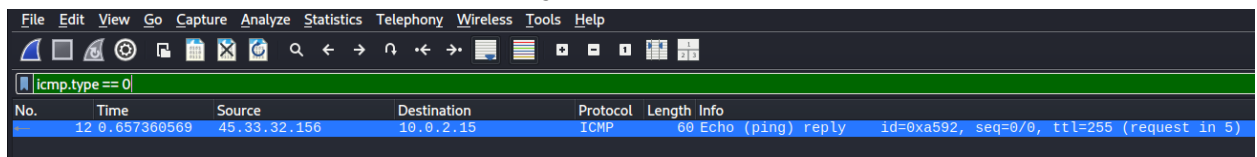
3. How many ICMP Echo Replies were received? (Use filter icmp.type == 0) Ans: 1



The screenshot shows the Wireshark interface with the filter 'icmp.type == 0' applied. A single packet is listed in the packet list pane.

| No. | Time        | Source       | Destination | Protocol | Length | Info                                                         |
|-----|-------------|--------------|-------------|----------|--------|--------------------------------------------------------------|
| 12  | 0.657360569 | 45.33.32.156 | 10.0.2.15   | ICMP     | 60     | Echo (ping) reply id=0xa592, seq=0/0, ttl=255 (request in 5) |

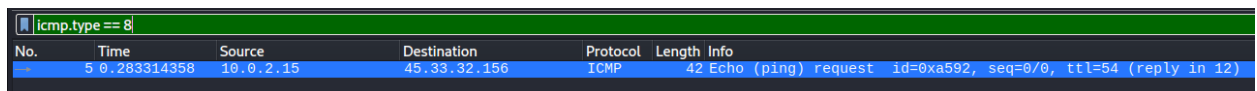
4. What is the TTL value in the Echo Reply packet? Ans: 255



The screenshot shows the Wireshark interface with the filter 'icmp.type == 0' applied. A single packet is listed in the packet list pane.

| No. | Time        | Source       | Destination | Protocol | Length | Info                                                         |
|-----|-------------|--------------|-------------|----------|--------|--------------------------------------------------------------|
| 12  | 0.657360569 | 45.33.32.156 | 10.0.2.15   | ICMP     | 60     | Echo (ping) reply id=0xa592, seq=0/0, ttl=255 (request in 5) |

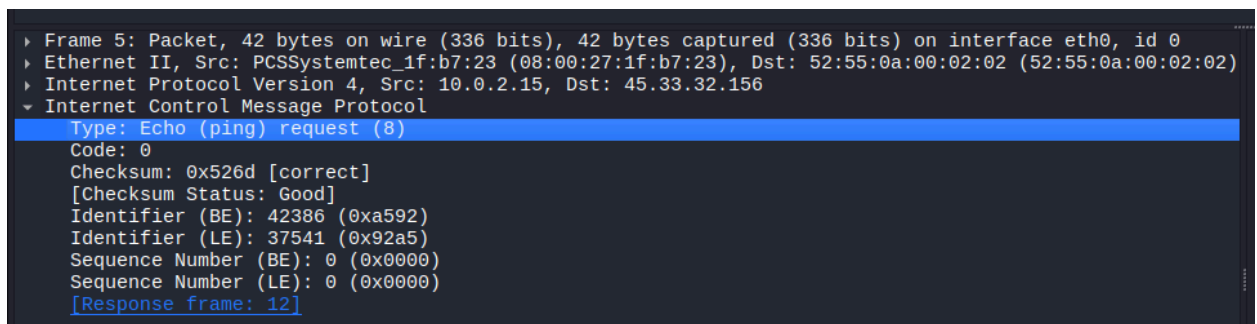
5. What is the source IP address in the Echo Request? Ans: 10.0.2.15 (My IP)



The screenshot shows the Wireshark interface with the filter 'icmp.type == 8' applied. A single packet is listed in the packet list pane.

| No. | Time        | Source    | Destination  | Protocol | Length | Info                                                         |
|-----|-------------|-----------|--------------|----------|--------|--------------------------------------------------------------|
| 5   | 0.283314358 | 10.0.2.15 | 45.33.32.156 | ICMP     | 42     | Echo (ping) request id=0xa592, seq=0/0, ttl=54 (reply in 12) |

6. What is the Identifier field value in the first Echo Request?



The screenshot shows the Wireshark interface with the packet details pane expanded for the first ICMP Echo request (packet 5). The fields are as follows:

| Field                | Value                   |
|----------------------|-------------------------|
| Type                 | Echo (ping) request (8) |
| Code                 | 0                       |
| Checksum             | 0x526d [correct]        |
| [Checksum Status]    | Good                    |
| Identifier (BE)      | 42386 (0xa592)          |
| Identifier (LE)      | 37541 (0x92a5)          |
| Sequence Number (BE) | 0 (0x0000)              |
| Sequence Number (LE) | 0 (0x0000)              |

## Task 2 - TCP Connect Scan

In a terminal, run a standard TCP connect scan: `nmap -sT -Pn -F <target>`

Apply the following display filters:

`tcp.flags.syn == 1 and tcp.flags.ack == 0` (Initial SYN only)

`tcp.flags.syn == 1 and tcp.flags.ack == 1` (SYN-ACK)

`tcp.flags.ack == 1 and tcp.flags.syn == 0 and ip.src == <scanner-ip>` (Final ACK of

handshake)

`tcp.flags.fin == 1 || tcp.flags.reset == 1`

Open Statistics > Flow Graph. Select 'TCP Flows'. You should see the SYN → SYN-ACK → ACK ladder diagram.

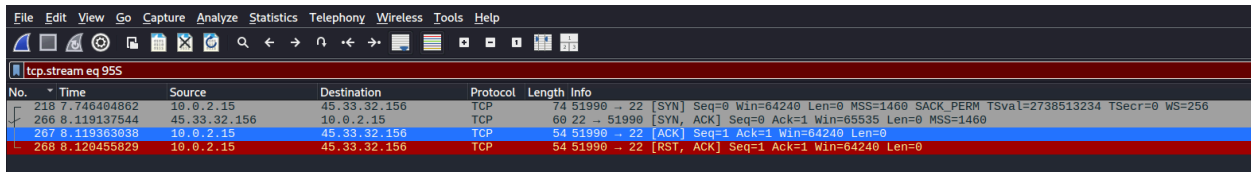
Analyze:

1. Identify one complete three-way handshake and record:

SYN packet number: 218

SYN/ACK packet number: 266

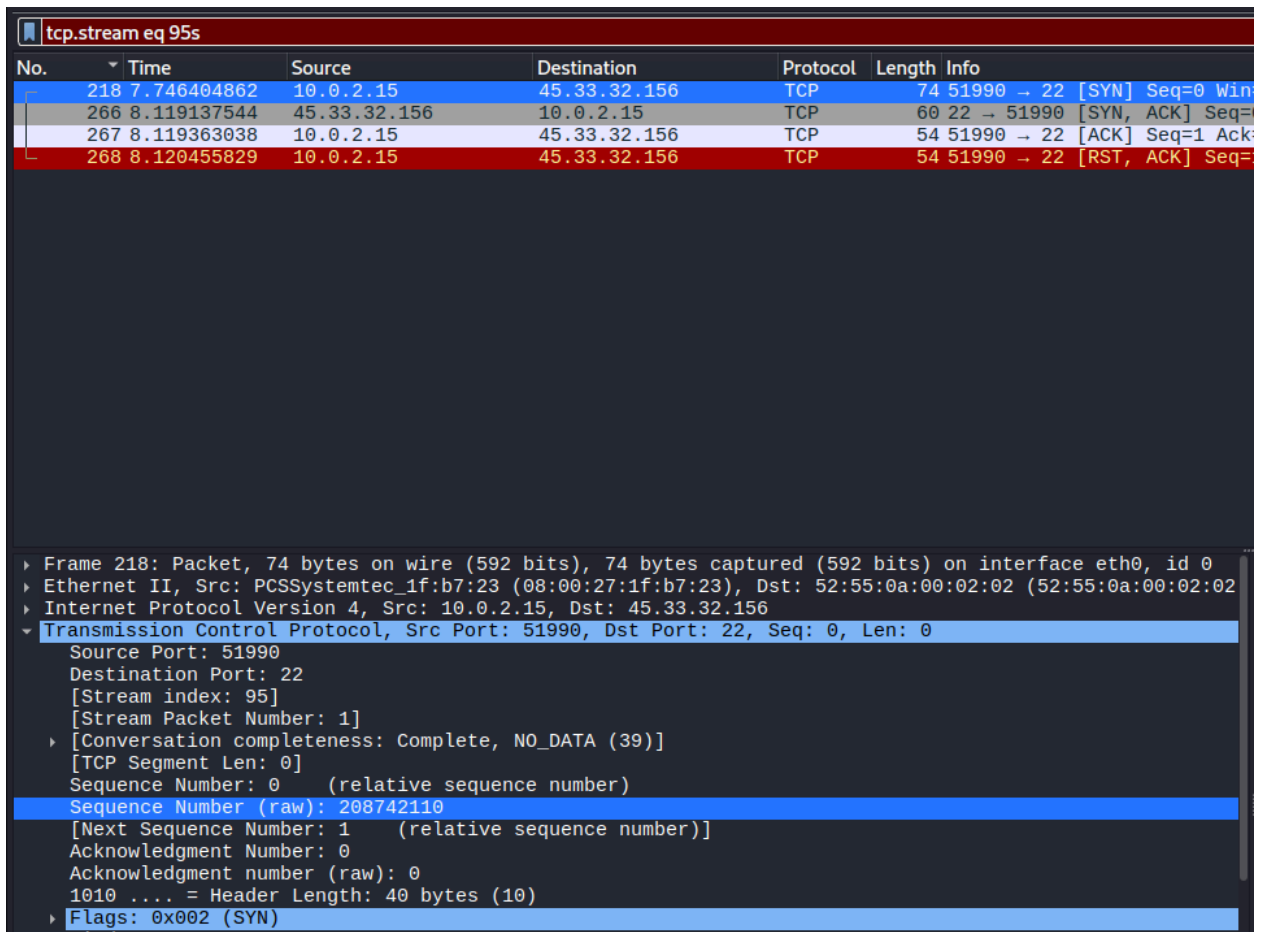
ACK packet number: 267



| No. | Time        | Source       | Destination  | Protocol | Length | Info                                                                                      |
|-----|-------------|--------------|--------------|----------|--------|-------------------------------------------------------------------------------------------|
| 218 | 7.746404862 | 10.0.2.15    | 45.33.32.156 | TCP      | 74     | 51990 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2738513234 TSecr=0 WS=256 |
| 266 | 8.119137544 | 45.33.32.156 | 10.0.2.15    | TCP      | 60     | 22 → 51990 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460                                |
| 267 | 8.119363038 | 10.0.2.15    | 45.33.32.156 | TCP      | 54     | 51990 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0                                              |
| 268 | 8.120455829 | 10.0.2.15    | 45.33.32.156 | TCP      | 54     | 51990 → 22 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0                                         |

2. For the selected handshake:

Client Initial Sequence Number: 208742110



| No. | Time        | Source       | Destination  | Protocol | Length | Info                                                                                      |
|-----|-------------|--------------|--------------|----------|--------|-------------------------------------------------------------------------------------------|
| 218 | 7.746404862 | 10.0.2.15    | 45.33.32.156 | TCP      | 74     | 51990 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2738513234 TSecr=0 WS=256 |
| 266 | 8.119137544 | 45.33.32.156 | 10.0.2.15    | TCP      | 60     | 22 → 51990 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460                                |
| 267 | 8.119363038 | 10.0.2.15    | 45.33.32.156 | TCP      | 54     | 51990 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0                                              |
| 268 | 8.120455829 | 10.0.2.15    | 45.33.32.156 | TCP      | 54     | 51990 → 22 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0                                         |

| Frame 218: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eth0, id 0    |
|---------------------------------------------------------------------------------------------------------|
| Ethernet II, Src: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02) |
| Internet Protocol Version 4, Src: 10.0.2.15, Dst: 45.33.32.156                                          |
| Transmission Control Protocol, Src Port: 51990, Dst Port: 22, Seq: 0, Len: 0                            |
| Source Port: 51990                                                                                      |
| Destination Port: 22                                                                                    |
| [Stream index: 95]                                                                                      |
| [Stream Packet Number: 1]                                                                               |
| [Conversation completeness: Complete, NO_DATA (39)]                                                     |
| [TCP Segment Len: 0]                                                                                    |
| Sequence Number: 0 (relative sequence number)                                                           |
| Sequence Number (raw): 208742110                                                                        |
| [Next Sequence Number: 1 (relative sequence number)]                                                    |
| Acknowledgment Number: 0                                                                                |
| Acknowledgment number (raw): 0                                                                          |
| 1010 .... = Header Length: 40 bytes (10)                                                                |
| Flags: 0x002 (SYN)                                                                                      |

Server Initial Sequence Number: 36416001

tcp.stream eq 95s

| No. | Time        | Source       | Destination  | Protocol | Length | Info                        |
|-----|-------------|--------------|--------------|----------|--------|-----------------------------|
| 218 | 7.746404862 | 10.0.2.15    | 45.33.32.156 | TCP      | 74     | 51990 → 22 [SYN] Seq=0 Win= |
| 266 | 8.119137544 | 45.33.32.156 | 10.0.2.15    | TCP      | 60     | 22 → 51990 [SYN, ACK] Seq=  |
| 267 | 8.119363038 | 10.0.2.15    | 45.33.32.156 | TCP      | 54     | 51990 → 22 [ACK] Seq=1 Ack= |
| 268 | 8.120455829 | 10.0.2.15    | 45.33.32.156 | TCP      | 54     | 51990 → 22 [RST, ACK] Seq=  |

▶ Frame 266: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0  
 ▶ Ethernet II, Src: 52:55:0a:00:02:02 (52:55:0a:00:02:02), Dst: PCSSystemtec\_1f:b7:23 (08:00:27:1f:b7:23)  
 ▶ Internet Protocol Version 4, Src: 45.33.32.156, Dst: 10.0.2.15  
 ▶ Transmission Control Protocol, Src Port: 22, Dst Port: 51990, Seq: 0, Ack: 1, Len: 0  
 Source Port: 22  
 Destination Port: 51990  
 [Stream index: 95]  
 [Stream Packet Number: 2]  
 ▶ [Conversation completeness: Complete, NO\_DATA (39)]  
 [TCP Segment Len: 0]  
 Sequence Number: 0 (relative sequence number)  
 Sequence Number (raw): 36416001  
 [Next Sequence Number: 1 (relative sequence number)]  
 Acknowledgment Number: 1 (relative ack number)  
 Acknowledgment number (raw): 208742111  
 0110 .... = Header Length: 24 bytes (6)  
 ▶ Flags: 0x012 (SYN, ACK)

### 3. How many connections were fully established? (Count completed SYN-SYN/ACK-ACK sequences)

tcp.flags.reset==0 and tcp.flags.ack==1 and ip.src==10.0.2.15

| No. | Time         | Source    | Destination  | Protocol | Length | Info                                           |
|-----|--------------|-----------|--------------|----------|--------|------------------------------------------------|
| 20  | 1.252625132  | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 51970 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0   |
| 49  | 3.331107521  | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 51986 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0   |
| 120 | 6.208443493  | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 51988 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0   |
| 267 | 8.119363038  | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 51990 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0   |
| 283 | 8.125626901  | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 49554 → 2000 [ACK] Seq=1 Ack=1 Win=64240 Len=0 |
| 289 | 8.126384764  | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 34006 → 5060 [ACK] Seq=1 Ack=1 Win=64240 Len=0 |
| 446 | 9.917445656  | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 49522 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0   |
| 555 | 11.581679007 | 10.0.2.15 | 45.33.32.156 | TCP      | 54     | 49532 → 22 [ACK] Seq=1 Ack=1 Win=64240 Len=0   |

8 fully established connections were there and 3 ports are open

tcp.flags.ack == 1 and tcp.flags.syn == 1

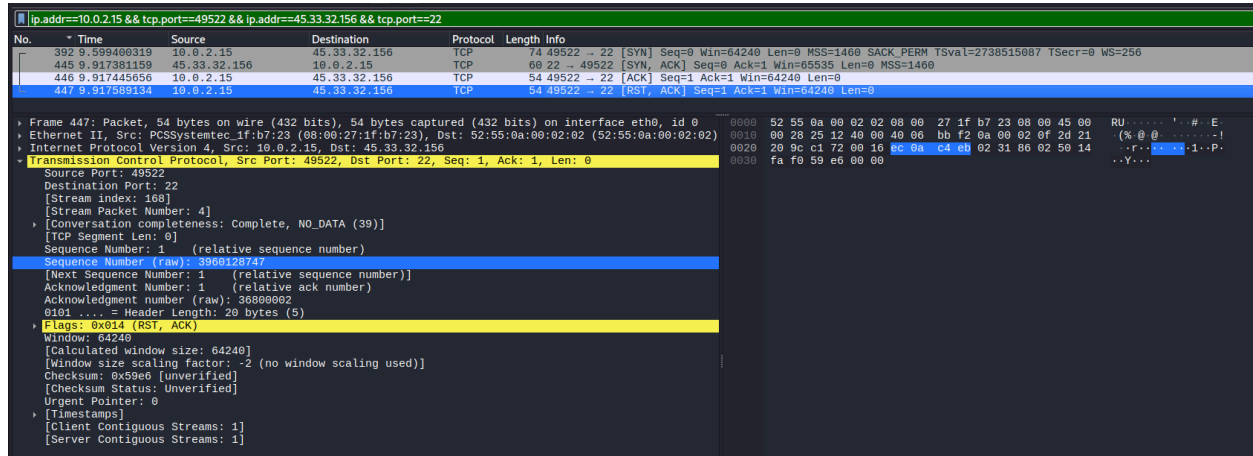
| No. | Time         | Source       | Destination | Protocol | Length | Info                                                         |
|-----|--------------|--------------|-------------|----------|--------|--------------------------------------------------------------|
| 19  | 1.252599888  | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 22 → 51970 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460   |
| 48  | 3.331093058  | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 22 → 51986 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460   |
| 119 | 6.208294175  | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 22 → 51988 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460   |
| 266 | 8.119137544  | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 22 → 51990 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460   |
| 281 | 8.125337890  | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 2000 → 49554 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 |
| 288 | 8.126369452  | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 5060 → 34006 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 |
| 445 | 9.917381159  | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 22 → 49522 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460   |
| 554 | 11.581173272 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 22 → 49532 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460   |

Wireshark: Conversations - test.pcapng

| Conversation Settings   | Ethernet - 1 | IPv4 - 1         | IPv6         | TCP - 8 | UDP        |
|-------------------------|--------------|------------------|--------------|---------|------------|
| Name resolution         | Address A    | Port A Address B | Port B       | Packets | Bytes      |
| Absolute start time     | 10.0.2.15    | 34006            | 45.33.32.156 | 5060    | 1 60 bytes |
| Display raw data        | 10.0.2.15    | 49532            | 45.33.32.156 | 22      | 1 60 bytes |
| Limit to display filter | 10.0.2.15    | 49554            | 45.33.32.156 | 2000    | 1 60 bytes |
| Copy                    | 10.0.2.15    | 51970            | 45.33.32.156 | 22      | 1 60 bytes |
| Follow Stream...        | 10.0.2.15    | 51988            | 45.33.32.156 | 22      | 1 60 bytes |
|                         | 10.0.2.15    | 51990            | 45.33.32.156 | 22      | 1 60 bytes |

#### 4. After connection establishment, does Nmap close the connection using FIN or RST?

**Ans:** Nmap will always close connections with the RST flag because it doesn't send any kind of data. In normal conversation after data sharing FIN flag is shared.

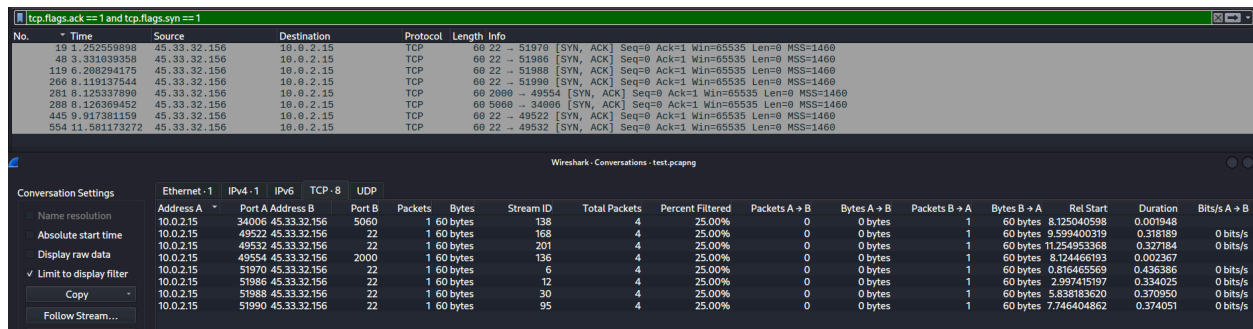


#### 5. How many ports were open and how many were closed?

3 ports are open

I haven't mentioned -F (100) so it will have 1000 ports. So answer would be

Closed ports = Total ports - open ports - filtered ports



**Ans:** Total open ports are 3

| tcp.flags.reset==1 and ip.src==45.33.32.156                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |              |             |          |        |                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|-------------|----------|--------|-----------------------------------------------------|
| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Time        | Source       | Destination | Protocol | Length | Info                                                |
| 75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.535412041 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 199 → 43888 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 76                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.535412612 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 3306 → 56984 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 77                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.568497376 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 5908 → 52292 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 78                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.606597935 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 1925 → 41636 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 79                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.647991695 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 111 → 33824 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.712906852 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 8080 → 38958 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 81                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.712907224 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 587 → 38510 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 82                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.712907250 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 1720 → 51302 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 83                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.978632474 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 993 → 40224 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4.990701086 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 25 → 34228 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0   |
| 250                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.857051498 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 8080 → 38904 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 251                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.868150814 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 113 → 58296 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 252                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.891610955 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 1720 → 51318 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 253                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.931979218 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 25 → 34238 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0   |
| 254                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.931979799 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 443 → 52802 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.931979862 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 1025 → 41650 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.942388146 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 110 → 54026 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 257                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.960135495 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 993 → 40236 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 258                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.960135046 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 111 → 33832 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 259                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.971587895 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 587 → 38514 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 260                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.995379325 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 995 → 42030 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 261                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7.995379677 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 5908 → 52306 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 262                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8.010062245 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 199 → 43904 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 263                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8.053584622 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 554 → 59426 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 264                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8.053585623 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 3306 → 56994 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| 265                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8.078875515 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 139 → 52604 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9.401578382 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 113 → 58912 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 420                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9.659875642 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 515 → 54176 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 421                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9.659876049 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 554 → 59430 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0  |
| 422                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9.729013007 | 45.33.32.156 | 10.0.2.15   | TCP      | 60     | 3306 → 56994 [RST, ACK] Seq=1 Ack=1 Win=65535 Len=0 |
| Frame 84: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0<br>Ethernet II, Src: 52:55:0a:00:02:02 (52:55:0a:00:02:02), Dst: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23)<br>Internet Protocol Version 4, Src: 45.33.32.156, Dst: 10.0.2.15<br>Transmission Control Protocol, Src Port: 25, Dst Port: 34228, Seq: 1, Ack: 1, Len: 0<br>Source Port: 25<br>Destination Port: 34228<br>[Stream index: 10]<br>[Stream Packet Number: 4]<br>[Conversation completeness: Incomplete (37)]<br>[TCP Segment Len: 0]<br>Sequence Number: 1 (relative sequence number)<br>Sequence Number (raw): 0<br>Next Sequence Number: 1 (relative sequence number) |             |              |             |          |        |                                                     |
| 0000 08 00 27 1f b7 23 52 55 0a 00 02 02 08 00 45 00 ...<br>0010 00 28 00 f1 00 00 40 06 20 14 2d 21 20 9c 0a 00 (...)<br>0020 02 0f 00 19 85 b4 00 00 00 00 26 a4 47 81 50 14 (...)<br>0030 ff ff 62 12 00 00 00 00 00 00 00 00 00 00 ...b...                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |              |             |          |        |                                                     |
| This shows the raw value of the sequence number (tcp.seq_raw), 4 bytes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |              |             |          |        | Packets: 599 - Displayed: 152 (25.4%)               |

Total packets = 152

Approx  $152/2 = 76$  are approx closed ports

Both are correct

## Task 3: FIN Scan

1. Generate FIN Scan Traffic by running:

`nmap -sF -Pn -F <target>`

2. Apply display filters like:

`tcp.flags.fin == 1`

`tcp.flags.reset == 1`

Analyze:

1. How many FIN-only packets were sent by the scanner? (Use FIN filter and count packets.) 100

| No. | Time        | Source    | Destination    | Protocol | Length | Info                                    |
|-----|-------------|-----------|----------------|----------|--------|-----------------------------------------|
| 9   | 0.577607352 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 3306 [FIN] Seq=1 Win=1024 Len=0 |
| 11  | 0.578788733 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 8888 [FIN] Seq=1 Win=1024 Len=0 |
| 13  | 0.579929093 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 111 [FIN] Seq=1 Win=1024 Len=0  |
| 15  | 0.581644711 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 25 [FIN] Seq=1 Win=1024 Len=0   |
| 17  | 0.582683209 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 445 [FIN] Seq=1 Win=1024 Len=0  |
| 19  | 0.583731259 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 3389 [FIN] Seq=1 Win=1024 Len=0 |
| 21  | 0.585132666 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 110 [FIN] Seq=1 Win=1024 Len=0  |
| 23  | 0.585999998 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 443 [FIN] Seq=1 Win=1024 Len=0  |
| 25  | 0.586869896 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 587 [FIN] Seq=1 Win=1024 Len=0  |
| 27  | 0.587731513 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 1720 [FIN] Seq=1 Win=1024 Len=0 |
| 29  | 0.588639101 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 993 [FIN] Seq=1 Win=1024 Len=0  |
| 31  | 0.589427366 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 113 [FIN] Seq=1 Win=1024 Len=0  |
| 33  | 0.590304485 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 554 [FIN] Seq=1 Win=1024 Len=0  |
| 35  | 0.591045139 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 135 [FIN] Seq=1 Win=1024 Len=0  |
| 37  | 0.591803901 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 995 [FIN] Seq=1 Win=1024 Len=0  |
| 39  | 0.592542276 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 139 [FIN] Seq=1 Win=1024 Len=0  |
| 41  | 0.593167695 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 80 [FIN] Seq=1 Win=1024 Len=0   |
| 43  | 0.593564828 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 23 [FIN] Seq=1 Win=1024 Len=0   |
| 45  | 0.594315997 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 5900 [FIN] Seq=1 Win=1024 Len=0 |
| 47  | 0.595059538 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 1723 [FIN] Seq=1 Win=1024 Len=0 |
| 49  | 0.595701659 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 22 [FIN] Seq=1 Win=1024 Len=0   |
| 51  | 0.596384790 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 21 [FIN] Seq=1 Win=1024 Len=0   |
| 53  | 0.597210350 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 8080 [FIN] Seq=1 Win=1024 Len=0 |
| 55  | 0.597876841 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 199 [FIN] Seq=1 Win=1024 Len=0  |
| 57  | 0.598521305 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 1025 [FIN] Seq=1 Win=1024 Len=0 |
| 59  | 0.599173057 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 143 [FIN] Seq=1 Win=1024 Len=0  |
| 61  | 0.599810737 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 53 [FIN] Seq=1 Win=1024 Len=0   |
| 63  | 0.600439573 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 1027 [FIN] Seq=1 Win=1024 Len=0 |
| 65  | 0.601187925 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 8089 [FIN] Seq=1 Win=1024 Len=0 |

Frame 9: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface eth0, id 0  
 Ethernet II, Src: PCSysstemec-1f:b7:23 (08:00:27:1f:b7:23), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)  
 Internet Protocol Version 4, Src: 10.0.2.15, Dst: 185.230.63.171  
 Transmission Control Protocol, Src Port: 34196, Dst Port: 3306, Seq: 1, Len: 0

## 2. What is the timestamp of the first FIN packet?

| No. | Time        | Source    | Destination    | Protocol | Length | Info                                    |
|-----|-------------|-----------|----------------|----------|--------|-----------------------------------------|
| 9   | 0.577607352 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 3306 [FIN] Seq=1 Win=1024 Len=0 |
| 11  | 0.578788733 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 8888 [FIN] Seq=1 Win=1024 Len=0 |
| 13  | 0.579929093 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 111 [FIN] Seq=1 Win=1024 Len=0  |
| 15  | 0.581644711 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 25 [FIN] Seq=1 Win=1024 Len=0   |
| 17  | 0.582683209 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 445 [FIN] Seq=1 Win=1024 Len=0  |
| 19  | 0.583731259 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 3389 [FIN] Seq=1 Win=1024 Len=0 |
| 21  | 0.585132666 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 110 [FIN] Seq=1 Win=1024 Len=0  |
| 23  | 0.585999998 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 443 [FIN] Seq=1 Win=1024 Len=0  |
| 25  | 0.586869896 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 587 [FIN] Seq=1 Win=1024 Len=0  |
| 27  | 0.587731513 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 1720 [FIN] Seq=1 Win=1024 Len=0 |
| 29  | 0.588639101 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 993 [FIN] Seq=1 Win=1024 Len=0  |
| 31  | 0.589427366 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 113 [FIN] Seq=1 Win=1024 Len=0  |
| 33  | 0.590304485 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 554 [FIN] Seq=1 Win=1024 Len=0  |
| 35  | 0.591045139 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 135 [FIN] Seq=1 Win=1024 Len=0  |
| 37  | 0.591803901 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 995 [FIN] Seq=1 Win=1024 Len=0  |

Frame 9: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface eth0, id 0  
 Section number: 1  
 Interface id: 0 (eth0)  
 Encapsulation type: Ethernet (1)  
 Arrival Time: Mar 3, 2026 05:30:56.510770816 EST  
 UTC Arrival Time: Mar 3, 2026 10:30:56.510770816 UTC  
 Epoch Arrival Time: 1772533856.510770816  
 [Time shift for this packet: 0.000000000 seconds]  
 [Time delta from previous captured frame: 7.223010 milliseconds]  
 [Time since reference or first frame: 577.607352 milliseconds]  
 Frame Number: 9  
 Frame Length: 54 bytes (432 bits)  
 Capture Length: 54 bytes (432 bits)  
 [Frame is marked: False]  
 [Frame is ignored: False]  
 [Protocols in frame: eth:ethertype:ip:tcp]  
 Character encoding: ASCII (0)  
 [Coloring Rule Name: TCP SYN/FIN]  
 [Coloring Rule String: tcp.flags & 0x02 || tcp.flags.fin == 1]

## 3. What is the timestamp of the last FIN packet?



| tcp.flags.fin == 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |           |                |          |        |                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|----------------|----------|--------|------------------------------------------|
| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Time        | Source    | Destination    | Protocol | Length | Info                                     |
| 179                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.632371955 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 81 [FIN] Seq=1 Win=1024 Len=0    |
| 181                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.633281296 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 1755 [FIN] Seq=1 Win=1024 Len=0  |
| 183                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.634015399 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 1433 [FIN] Seq=1 Win=1024 Len=0  |
| 185                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.634544057 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 9100 [FIN] Seq=1 Win=1024 Len=0  |
| 187                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.635055881 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 49152 [FIN] Seq=1 Win=1024 Len=0 |
| 189                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.635465299 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 5000 [FIN] Seq=1 Win=1024 Len=0  |
| 191                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.635964805 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 37 [FIN] Seq=1 Win=1024 Len=0    |
| 193                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.636757365 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 8000 [FIN] Seq=1 Win=1024 Len=0  |
| 195                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.637259509 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 5051 [FIN] Seq=1 Win=1024 Len=0  |
| 197                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.637747332 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 5631 [FIN] Seq=1 Win=1024 Len=0  |
| 199                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.638243796 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 8008 [FIN] Seq=1 Win=1024 Len=0  |
| 201                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.638730411 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 5101 [FIN] Seq=1 Win=1024 Len=0  |
| 203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.639234969 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 5009 [FIN] Seq=1 Win=1024 Len=0  |
| 205                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.639722567 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 9999 [FIN] Seq=1 Win=1024 Len=0  |
| 207                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0.640245428 | 10.0.2.15 | 185.230.63.171 | TCP      | 54     | 34196 → 49157 [FIN] Seq=1 Win=1024 Len=0 |
| ▾ Frame 207: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface eth0, id 0<br>Section number: 1<br>▸ Interface id: 0 (eth0)<br>Encapsulation type: Ethernet (1)<br>Arrival Time: Mar 3, 2026 05:30:56.573408892 EST<br>UTC Arrival Time: Mar 3, 2026 10:30:56.573408892 UTC<br>Epoch Arrival Time: 1772533856.573408892<br>[Time shift for this packet: 0.000000000 seconds]<br>[Time delta from previous captured frame: 293.998 microseconds]<br>[Time delta from previous displayed frame: 522.861 microseconds]<br>[Time since reference or first frame: 640.245428 milliseconds]<br>Frame Number: 207<br>Frame Length: 54 bytes (432 bits)<br>Capture Length: 54 bytes (432 bits)<br>[Frame is marked: False]<br>[Frame is ignored: False]<br>[Protocols in frame: eth:ethertype:ip:tcp]<br>Character encoding: ASCII (0)<br>[Coloring Rule Name: TCP SYN/FIN]<br>[Coloring Rule String: tcp.flags & 0x02    tcp.flags.fin == 1]<br>▾ Ethernet II, Src: PCSSystemtec_if:b7:23 (08:00:27:1f:b7:23), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02) |             |           |                |          |        |                                          |

4. What is the total scan duration? (Last FIN – First FIN)

5. How many RST packets were received from the target?

| tcp.flags.reset == 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                |             |          |        |                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------|-------------|----------|--------|--------------------------------------------------|
| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Time        | Source         | Destination | Protocol | Length | Info                                             |
| 180                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.632572819 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 81 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0    |
| 182                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.633489217 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 1755 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 184                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.634397799 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 1433 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 186                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.634749805 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 9100 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 188                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.635318278 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 49152 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0 |
| 190                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.635605370 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 5000 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 192                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.636257806 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 37 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0    |
| 194                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.636959466 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 8000 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 196                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.637453046 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 5051 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 198                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.637940093 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 5631 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.638437146 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 8008 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 202                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.638925289 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 5101 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 204                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.639428465 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 5009 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 206                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.639951430 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 9999 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0  |
| 208                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.640438420 | 185.230.63.171 | 10.0.2.15   | TCP      | 60     | 49157 → 34196 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0 |
| ▾ Frame 206: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0<br>Section number: 1<br>▸ Interface id: 0 (eth0)<br>Encapsulation type: Ethernet (1)<br>Arrival Time: Mar 3, 2026 05:30:56.573114894 EST<br>UTC Arrival Time: Mar 3, 2026 10:30:56.573114894 UTC<br>Epoch Arrival Time: 1772533856.573114894<br>[Time shift for this packet: 0.000000000 seconds]<br>[Time delta from previous captured frame: 228.863 microseconds]<br>[Time delta from previous displayed frame: 522.965 microseconds]<br>[Time since reference or first frame: 639.951430 milliseconds]<br>Frame Number: 206<br>Frame Length: 60 bytes (480 bits)<br>Capture Length: 60 bytes (480 bits)<br>[Frame is marked: False]<br>[Frame is ignored: False]<br>[Protocols in frame: eth:ethertype:ip:tcp]<br>Character encoding: ASCII (0)<br>[Coloring Rule Name: TCP RST]<br>[Coloring Rule String: tcp.flags.reset eq 1]<br>▾ Ethernet II, Src: 52:55:0a:00:02:02 (52:55:0a:00:02:02), Dst: PCSSystemtec_if:b7:23 (08:00:27:1f:b7:23)<br>▾ Destination: PCSSystemtec_if:b7:23 (08:00:27:1f:b7:23)<br>....:0 .... = IG bit: Globally unique address (factory default)<br>....:0 .... = IG bit: Individual address (unicast)<br>▾ Source: 52:55:0a:00:02:02 (52:55:0a:00:02:02)<br>....:1 .... = LG bit: Locally administered address (this is NOT the factory default)<br>....:0 .... = IG bit: Individual address (unicast)<br>Type: 0x00000000 |             |                |             |          |        |                                                  |

6. How many FIN packets did NOT receive any response?

# Part B: PCAP

Use Wireshark to analyze the pcap provided and answer the following questions:

## 1. Identify the scanning technique used.

| tcp.flags.reset==0 and tcp.flags.ack==1 |          |              |              |          |        |                                                              |
|-----------------------------------------|----------|--------------|--------------|----------|--------|--------------------------------------------------------------|
| No.                                     | Time     | Source       | Destination  | Protocol | Length | Info                                                         |
| 190                                     | 0.039475 | 192.168.1.25 | 192.168.1.10 | TCP      | 60     | 445 → 35503 [SYN, ACK] Seq=0 Ack=1 Win=17520 Len=0 MSS=1460  |
| 482                                     | 0.082828 | 192.168.1.25 | 192.168.1.10 | TCP      | 60     | 5000 → 35503 [SYN, ACK] Seq=0 Ack=1 Win=17520 Len=0 MSS=1460 |
| 938                                     | 1.167638 | 192.168.1.25 | 192.168.1.10 | TCP      | 60     | 139 → 35503 [SYN, ACK] Seq=0 Ack=1 Win=17520 Len=0 MSS=1460  |
| 1620                                    | 1.206473 | 192.168.1.25 | 192.168.1.10 | TCP      | 60     | 135 → 35503 [SYN, ACK] Seq=0 Ack=1 Win=17520 Len=0 MSS=1460  |
| 1731                                    | 1.210195 | 192.168.1.25 | 192.168.1.10 | TCP      | 60     | 1025 → 35503 [SYN, ACK] Seq=0 Ack=1 Win=17520 Len=0 MSS=1460 |

| tcp.flags.reset==1 and tcp.flags.ack==0 |          |              |              |          |        |                                      |
|-----------------------------------------|----------|--------------|--------------|----------|--------|--------------------------------------|
| No.                                     | Time     | Source       | Destination  | Protocol | Length | Info                                 |
| 195                                     | 0.039720 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 445 [RST] Seq=1 Win=0 Len=0  |
| 483                                     | 0.082849 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 5000 [RST] Seq=1 Win=0 Len=0 |
| 939                                     | 1.167656 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 139 [RST] Seq=1 Win=0 Len=0  |
| 1621                                    | 1.206491 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 135 [RST] Seq=1 Win=0 Len=0  |
| 1732                                    | 1.210206 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 1025 [RST] Seq=1 Win=0 Len=0 |

When you follow the tcp stream of each packet you get syn, syn/ack and rst.

## TCP SYN Scan (Half-Open Scan)

Pattern:

SYN →

SYN,ACK ← (if port open)

RST → (scanner immediately resets instead of completing handshake)

## TCP Connect Scan (Full Connect Scan)

Pattern:

SYN →

SYN,ACK ←

ACK →

Connection fully established

Then closed (FIN or RST)

You will see a complete 3-way handshake. This is stealth scan

## 2. Identify the IP address of the attacker. Ans: 192.168.1.10

| No. | Time     | Source       | Destination  | Protocol | Length | Info                                             |
|-----|----------|--------------|--------------|----------|--------|--------------------------------------------------|
| 1   | 0.000000 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 1723 [SYN] Seq=0 Win=2048 Len=0 MSS=1460 |
| 2   | 0.000078 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 443 [SYN] Seq=0 Win=2048 Len=0 MSS=1460  |
| 3   | 0.000122 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 23 [SYN] Seq=0 Win=4096 Len=0 MSS=1460   |
| 4   | 0.000165 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 22 [SYN] Seq=0 Win=3072 Len=0 MSS=1460   |
| 5   | 0.000208 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460 |
| 6   | 0.000249 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460  |
| 7   | 0.000290 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 113 [SYN] Seq=0 Win=3072 Len=0 MSS=1460  |
| 8   | 0.000333 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 21 [SYN] Seq=0 Win=3072 Len=0 MSS=1460   |
| 9   | 0.000376 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 256 [SYN] Seq=0 Win=4096 Len=0 MSS=1460  |

## 3. Identify the IP address of the victim. Ans: 192.168.1.25

| No. | Time     | Source       | Destination  | Protocol | Length | Info                                             |
|-----|----------|--------------|--------------|----------|--------|--------------------------------------------------|
| 1   | 0.000000 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 1723 [SYN] Seq=0 Win=2048 Len=0 MSS=1460 |
| 2   | 0.000078 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 443 [SYN] Seq=0 Win=2048 Len=0 MSS=1460  |
| 3   | 0.000122 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 23 [SYN] Seq=0 Win=4096 Len=0 MSS=1460   |
| 4   | 0.000165 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 22 [SYN] Seq=0 Win=3072 Len=0 MSS=1460   |
| 5   | 0.000208 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460 |
| 6   | 0.000249 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460  |
| 7   | 0.000290 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 113 [SYN] Seq=0 Win=3072 Len=0 MSS=1460  |
| 8   | 0.000333 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 21 [SYN] Seq=0 Win=3072 Len=0 MSS=1460   |
| 9   | 0.000376 | 192.168.1.10 | 192.168.1.25 | TCP      | 58     | 35503 → 256 [SYN] Seq=0 Win=4096 Len=0 MSS=1460  |

#### 4. Determine the range of ports that were scanned.

1 - 65389

First filter traffic to the victim: ip.addr == <target\_IP>

Show Only Scan Packets

If it's a SYN scan (most common), use: tcp.flags.syn == 1 and tcp.flags.ack == 0

Wireshark - Conversations - sample3 (10).pcap

| Name resolution               | Address A    | Port A | Address B    | Port B | Packets | Bytes     | Stream ID | Total Packets | Percent Filtered | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel. Start | Duration | Bits/s A → B |
|-------------------------------|--------------|--------|--------------|--------|---------|-----------|-----------|---------------|------------------|---------------|-------------|---------------|-------------|------------|----------|--------------|
| Absolute start time           | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 1       | 58 bytes  | 558       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.174983   | 0.0005   |              |
| Display raw data              | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 4       | 158 bytes | 16        | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.001424   | 0.0009   |              |
| Limit to display filter       | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 6       | 158 bytes | 930       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.228353   | 0.0010   |              |
| Copy                          | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 7       | 158 bytes | 330       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.126261   | 0.0013   |              |
| Follow Stream...              | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 9       | 158 bytes | 715       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.185309   | 0.0004   |              |
| Graph...                      | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 13      | 158 bytes | 35        | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.002317   | 0.0334   | 13 kbps      |
| I/O Graphs                    | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 17      | 158 bytes | 649       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.181032   | 0.0010   |              |
| Protocol                      | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 22      | 158 bytes | 481       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.168944   | 0.0007   |              |
| Bluetooth                     | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 19      | 158 bytes | 913       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.221784   | 0.0057   | 81 kbps      |
| BPV                           | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 20      | 158 bytes | 909       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.221627   | 0.0053   | 87 kbps      |
| Filter list for specific type | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 21      | 158 bytes | 7         | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.000333   | 0.0006   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 22      | 158 bytes | 3         | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.000165   | 0.0005   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 23      | 158 bytes | 2         | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.000122   | 0.0005   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 24      | 158 bytes | 409       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.162439   | 0.0006   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 25      | 158 bytes | 9         | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.000418   | 0.0007   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 26      | 158 bytes | 757       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.187916   | 0.0013   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 30      | 158 bytes | 607       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.178041   | 0.0011   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 31      | 158 bytes | 424       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.164555   | 0.0006   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 33      | 158 bytes | 637       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.180473   | 0.0011   |              |

Wireshark - Conversations - sample3 (10).pcap

| Name resolution               | Address A    | Port A | Address B    | Port B | Packets | Bytes     | Stream ID | Total Packets | Percent Filtered | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel. Start | Duration | Bits/s A → B |
|-------------------------------|--------------|--------|--------------|--------|---------|-----------|-----------|---------------|------------------|---------------|-------------|---------------|-------------|------------|----------|--------------|
| Absolute start time           | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 1       | 58 bytes  | 891       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.220674   | 0.0059   | 79 kbps      |
| Display raw data              | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 4       | 158 bytes | 855       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.209663   | 0.0010   |              |
| Limit to display filter       | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 6       | 158 bytes | 782       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.204791   | 0.0005   |              |
| Copy                          | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 7       | 158 bytes | 464       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.167913   | 0.0001   |              |
| Follow Stream...              | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 9       | 158 bytes | 5738      | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.168438   | 0.0004   |              |
| Graph...                      | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 13      | 158 bytes | 7787      | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.162481   | 0.0006   |              |
| I/O Graphs                    | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 17      | 158 bytes | 58080     | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.207574   | 0.0006   |              |
| Protocol                      | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 22      | 158 bytes | 927       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.228201   | 0.0009   |              |
| Bluetooth                     | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 19      | 158 bytes | 71        | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.018075   | 0.0199   | 23 kbps      |
| BPV                           | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 20      | 158 bytes | 405       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.162285   | 0.0004   |              |
| Filter list for specific type | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 21      | 158 bytes | 866       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.210127   | 0.0009   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 22      | 158 bytes | 926       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.228153   | 0.0008   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 23      | 158 bytes | 105       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.048507   | 0.0007   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 24      | 158 bytes | 740       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.187743   | 0.0003   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 31      | 158 bytes | 331       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.126303   | 0.0013   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 33      | 158 bytes | 974       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.236787   | 0.0033   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 30      | 158 bytes | 883       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.220172   | 0.0061   | 76 kbps      |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 33      | 158 bytes | 939       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.228736   | 0.0012   |              |

#### 5. Determine how many ports were closed.

Wireshark - Conversations - sample3 (10).pcap

| Name resolution               | Address A    | Port A | Address B    | Port B | Packets | Bytes     | Stream ID | Total Packets | Percent Filtered | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel. Start | Duration | Bits/s A → B |
|-------------------------------|--------------|--------|--------------|--------|---------|-----------|-----------|---------------|------------------|---------------|-------------|---------------|-------------|------------|----------|--------------|
| Absolute start time           | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 1       | 58 bytes  | 891       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.220674   | 0.0059   | 79 kbps      |
| Display raw data              | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 4       | 158 bytes | 855       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.209663   | 0.0010   |              |
| Limit to display filter       | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 6       | 158 bytes | 782       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.204791   | 0.0005   |              |
| Copy                          | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 7       | 158 bytes | 464       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.167913   | 0.0001   |              |
| Follow Stream...              | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 9       | 158 bytes | 5738      | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.168438   | 0.0004   |              |
| Graph...                      | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 13      | 158 bytes | 7787      | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.162481   | 0.0006   |              |
| I/O Graphs                    | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 17      | 158 bytes | 58080     | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.207574   | 0.0006   |              |
| Protocol                      | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 22      | 158 bytes | 927       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.228201   | 0.0009   |              |
| Bluetooth                     | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 19      | 158 bytes | 71        | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.018075   | 0.0199   | 23 kbps      |
| BPV                           | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 20      | 158 bytes | 405       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.162285   | 0.0004   |              |
| Filter list for specific type | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 21      | 158 bytes | 866       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.210127   | 0.0009   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 22      | 158 bytes | 926       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.228153   | 0.0008   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 23      | 158 bytes | 105       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.048507   | 0.0007   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 24      | 158 bytes | 740       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.187743   | 0.0003   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 31      | 158 bytes | 331       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 0.126303   | 0.0013   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 33      | 158 bytes | 974       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.236787   | 0.0033   |              |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 30      | 158 bytes | 883       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.220172   | 0.0061   | 76 kbps      |
|                               | 192.168.1.10 | 35503  | 192.168.1.25 | 41511  | 33      | 158 bytes | 939       | 2             | 50.00%           | 1             | 58 bytes    | 0             | 0 bytes     | 1.228736   | 0.0012   |              |

| Flags    | Meaning in SYN Scan |
|----------|---------------------|
| SYN, ACK | Port OPEN           |
| RST, ACK | Port CLOSED         |
| No reply | Port FILTERED       |

## 6. Determine how many ports were open, and list the open port numbers.

| tcp.flags.reset==1 and tcp.flags.ack==0 |          |              |              |          |        |                                      |  |  |  |
|-----------------------------------------|----------|--------------|--------------|----------|--------|--------------------------------------|--|--|--|
| No.                                     | Time     | Source       | Destination  | Protocol | Length | Info                                 |  |  |  |
| 195                                     | 0.039720 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 445 [RST] Seq=1 Win=0 Len=0  |  |  |  |
| 483                                     | 0.062946 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 5000 [RST] Seq=1 Win=0 Len=0 |  |  |  |
| 939                                     | 1.107656 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 139 [RST] Seq=1 Win=0 Len=0  |  |  |  |
| 1621                                    | 1.206491 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 135 [RST] Seq=1 Win=0 Len=0  |  |  |  |
| 1732                                    | 1.210206 | 192.168.1.10 | 192.168.1.25 | TCP      | 54     | 35503 → 1025 [RST] Seq=1 Win=0 Len=0 |  |  |  |

| Wireshark - Conversations - sample3 (1)(2).pcap |              |          |              |          |     |          |          |           |               |
|-------------------------------------------------|--------------|----------|--------------|----------|-----|----------|----------|-----------|---------------|
| Conversation Settings                           | Ethernet - 1 | IPv4 - 1 | IPv6         | TCP - 5  | UDP | Packets  | Bytes    | Stream ID | Total Packets |
| Name resolution                                 | 192.168.1.10 | 35503    | 192.168.1.25 | Port B * | 135 | 1        | 54 bytes | 792       | 3             |
| Absolute start time                             | 192.168.1.10 | 35503    | 192.168.1.25 | 139      | 1   | 54 bytes | 460      | 3         | 33.33%        |
| Display raw data                                | 192.168.1.10 | 35503    | 192.168.1.25 | 445      | 1   | 54 bytes | 92       | 3         | 33.33%        |
| Limit to display filter                         | 192.168.1.10 | 35503    | 192.168.1.25 | 1025     | 1   | 54 bytes | 847      | 3         | 33.33%        |
|                                                 |              |          |              | 5000     | 1   | 54 bytes | 233      | 3         | 33.33%        |

**Ans:** Ports 135, 139, 445, 1025, 5000 are open, that's why they can get the response of syn/ack

| Scan Type                 | Packet Sent by Scanner | If Port is OPEN                     | If Port is CLOSED | If Port is FILTERED            |
|---------------------------|------------------------|-------------------------------------|-------------------|--------------------------------|
| <b>SYN (Stealth) Scan</b> | SYN                    | SYN, ACK                            | RST, ACK          | No response / ICMP unreachable |
| <b>Connect Scan</b>       | SYN                    | SYN, ACK → full handshake completes | RST, ACK          | No response / ICMP unreachable |
| <b>FIN Scan</b>           | FIN                    | No response                         | RST               | No response                    |
| <b>NULL Scan</b>          | No flags               | No response                         | RST               | No response                    |
| <b>XMAS Scan</b>          | FIN + PSH + URG        | No response                         | RST               | No response                    |
| <b>ACK Scan</b>           | ACK                    | RST                                 | RST               | No response / ICMP unreachable |
| <b>Window Scan</b>        | ACK                    | RST (non-zero window)               | RST (zero window) | No response                    |

If you see:

Many SYNs → likely SYN scan

Only FIN packets → FIN scan

No flags set → NULL scan

FIN+PSH+URG → XMAS scan

Only ACK packets → ACK scan